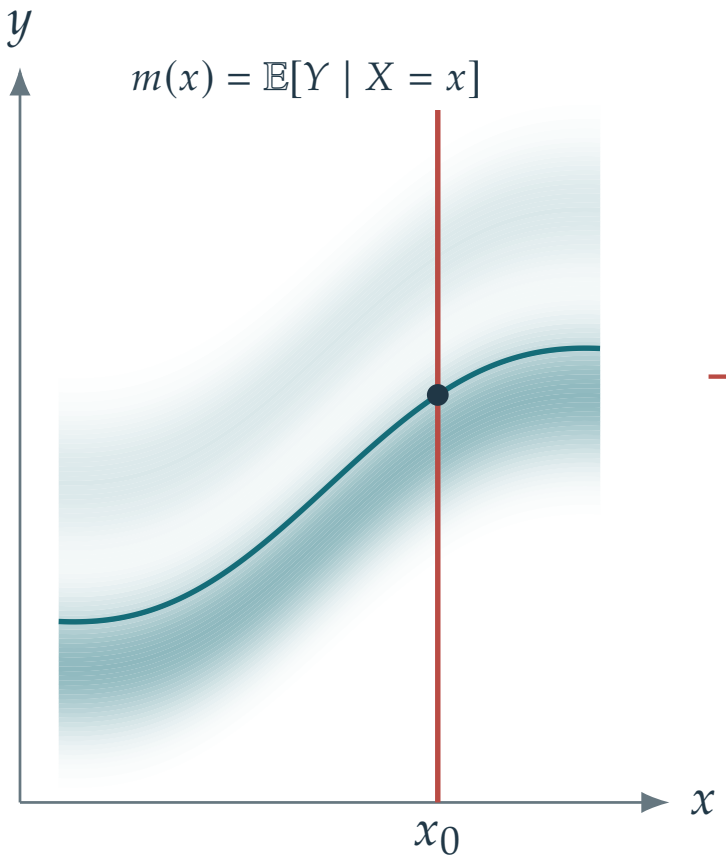
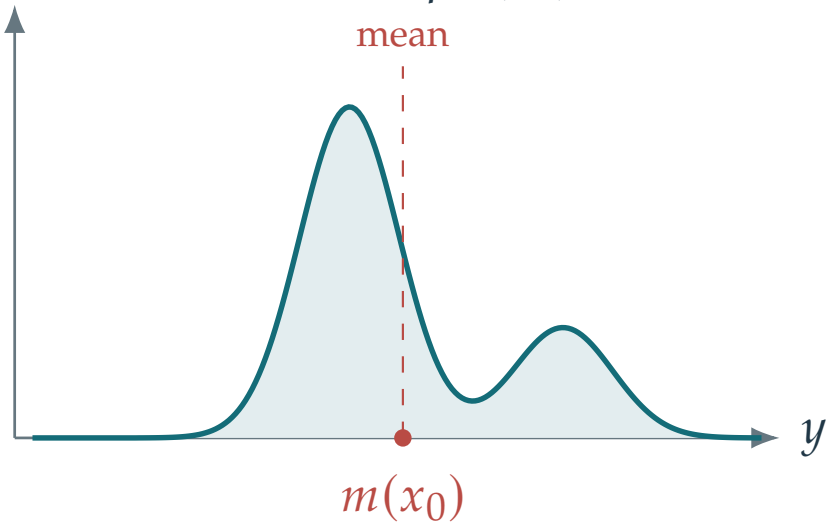


Fix $x = x_0$
and normalize

$$p(y \mid x_0) = \frac{\pi(x_0, y)}{p_X(x_0)}$$



$$m(x_0) = \int y\, p(y \mid x_0)\, \mathrm{d}y$$

Darker shading: higher joint density.

The mean need not be the most likely output.